

ECON602 TA Notes: Inventory

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This note discusses Blinder and Fisher (1981, JME). The paper is accessible on ELMS.

Introducing inventories into a firm's decision-making process is similar to allowing consumers to save. It gives the firm a way to allocate resources intertemporally and prepare for unforeseen contingencies (e.g., avoiding stockouts when there is an abrupt shortage).

Setup

Firm i faces downward-sloping demand:

$$P_{it} = v_{it}P_tD(X_{it}) \quad (1)$$

where P_{it} is the firm's nominal price, v_{it} is an idiosyncratic shock, P_t is the general price level, and $D(X_{it})$ is a demand function in which X_{it} represents sales. We assume that D is downward-sloping: $D'(\cdot) < 0$.

The (real) revenue of the firm can therefore be written as

$$R_{it} = \frac{P_{it}X_{it}}{P_t} = v_{it}D(X_{it})X_{it} \quad (2)$$

We require that R is increasing and concave in X_{it} .

Firm i faces costs

$$C_{it} = P_t c(Y_{it}), \quad c' > 0, \quad c'' > 0 \quad (3)$$

so nominal costs equal the general price level times real costs. The cost function is increasing and convex. Note that production Y_{it} can differ from sales X_{it} .

Let N_{it} denote the inventory of firm i at the beginning of period t . It evolves according to

$$N_{i,t+1} = N_{it} + Y_{it} - X_{it} \quad (4)$$

so next period's inventory stock equals today's inventory plus production minus sales. Note that inventory N_{it} can take negative numbers, which can be interpreted as a queue of unfilled orders or stockouts.

It is assumed that there is a cost $B(N_{it})$ involved in holding inventory. The cost function $B(\cdot)$ is assumed to be convex and

$$B'(N_{it}) < 0 \text{ if } N_{it} < 0 \text{ and } B'(N_{it}) > 0 \text{ otherwise.}$$

This captures the fact that both holding inventories and experiencing stockouts are costly for the firm.

Firm's Decision Problem

The firm chooses production Y_{it} , sales X_{it} , and next period's inventory $N_{i,t+1}$ to maximize lifetime profits:

$$\mathbb{E}_0 \sum_{t=0}^{\infty} (1+r)^{-t} \left[R_{it} - \frac{C_{it}}{P_t} - B(N_{it}) \right] \quad (5)$$

We can write the Bellman equation by following the same steps used last semester. First, identify the state and control variables. There is one endogenous state, inventory N , and one exogenous state, the idiosyncratic shock v_{it} ; the value function therefore takes these two arguments. The controls are production Y and sales X . Hence, the Bellman equation is

$$V(N; v) = \max_{X, Y, N'} vD(X)X - c(Y) - B(N) + \frac{1}{1+r} \mathbb{E}[V(N'; v')] \quad (6)$$

subject to

$$N' = N + Y - X$$

We follow the standard procedure to solve this Bellman equation. Substituting the law of motion for inventory into the right-hand side and eliminating N' as a control yields two FOCs for X and Y , respectively:

$$vD'(X)X + vD(x) - \frac{1}{1+r} \mathbb{E}[V_N(N'; v')] = 0 \quad (7)$$

and

$$-c'(Y) + \frac{1}{1+r} \mathbb{E}[V_N(N'; v')] = 0 \quad (8)$$

The envelope condition is

$$V_N(N; v) = -B'(N) + \frac{1}{1+r} \mathbb{E}[V_N(N'; v')] \quad (9)$$

Implications

Here are several observations:

- Combining (7) and (8), we get

$$vD'(X)X + vD(X) = c'(Y) \quad (10)$$

which says, roughly, that marginal revenue equals marginal cost.

- Combining (7) with the envelope condition (9):

$$vD'(X)X + vD(X) - B'(N) = V_N(N; v)$$

and hence

$$vD'(X)X + vD(X) = \frac{1}{1+r} \mathbb{E}[v'D'(X')X' + v'D(X') - B'(N')] \quad (11)$$

which is an Euler equation for sales and inventory. A similar Euler equation can be derived for output and inventory as well:

$$c'(Y) - B'(N) = V_N(N; v)$$

and hence

$$c'(Y) = \frac{1}{1+r} \mathbb{E}[c'(Y') - B'(N')] \quad (12)$$

In these equations, $B'(N')$ can be taken outside the expectation because N' is determined in the current period.

- Our goal is to illustrate the conditions under which inventories generate persistence in output—that is, how tomorrow's production Y' is affected by a temporary shock to v today. To show this, first apply the implicit function theorem to (10):

$$v(D''(X)X + D'(X) + D'(X))\frac{\partial X}{\partial v} + (D'(X)X + D(X)) = c''(Y)\frac{\partial Y}{\partial v}$$

which implies that

$$\frac{\partial X}{\partial v} = -\frac{D'(X)X + D(X)}{v(D''(X)X + D'(X) + D'(X))} + \frac{c''(Y)}{v(D''(X)X + D'(X) + D'(X))} \frac{\partial Y}{\partial v}$$

so that

$$\frac{\partial Y}{\partial v} - \frac{\partial X}{\partial v} = \left(1 - \frac{c''(Y)}{v(D''(X)X + D'(X) + D'(X))}\right) \frac{\partial Y}{\partial v} + \frac{D'(X)X + D(X)}{v(D''(X)X + D'(X) + D'(X))}$$

Similarly, apply the implicit function theorem to (12):

$$c''(Y) \frac{\partial Y}{\partial v} = \frac{1}{1+r} \mathbb{E} \left[c''(Y') \frac{\partial Y'}{\partial v} \right] - \frac{1}{1+r} B''(N + Y - X) \left[\frac{\partial Y}{\partial v} - \frac{\partial X}{\partial v} \right]$$

which implies that

$$\begin{aligned} \mathbb{E} \left[\frac{\partial Y'}{\partial v} \right] &= (1+r) \frac{\partial Y}{\partial v} + B''(N') \left[\frac{\partial Y}{\partial v} - \frac{\partial X}{\partial v} \right] \\ &= (1+r) \frac{\partial Y}{\partial v} + B''(N') \left[\left(1 - \frac{c''(Y)}{v(D''(X)X + 2D'(X))}\right) \frac{\partial Y}{\partial v} + \frac{D'(X)X + D(X)}{v(D''(X)X + 2D'(X))} \right] \\ &= \left[1 + r + B''(N') \left(1 - \frac{c''(Y)}{v(D''(X)X + 2D'(X))}\right) \right] \frac{\partial Y}{\partial v} + B''(N') \frac{D'(X)X + D(X)}{v(D''(X)X + 2D'(X))} \end{aligned} \tag{13}$$

Since we assumed that $c''(\cdot) > 0$, $B''(\cdot) > 0$, and

$$D'(X)X + D(X) > 0 \text{ and } D''(X)X + D'(X) + D'(X) < 0$$

that is, the revenue function is increasing and concave, the coefficient on $\frac{\partial Y}{\partial v}$ is strictly positive. In other words, variation in Y' caused by today's shock v partly reflects the variation in Y generated by that shock.